

Design technology Standard (practice) level Paper 1

Set A

1 hour**Instructions to candidates**

- Do not open this examination paper until instructed to do so.
- Answer all questions.
- A calculator is required for this paper.
- For each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.
- The maximum mark for the examination paper is [30 marks].

This is a practice paper by Mr. K, and may differ from official IB conventions.

1. The circular economy keeps materials in use for as long as possible instead of discarding them after a single life.

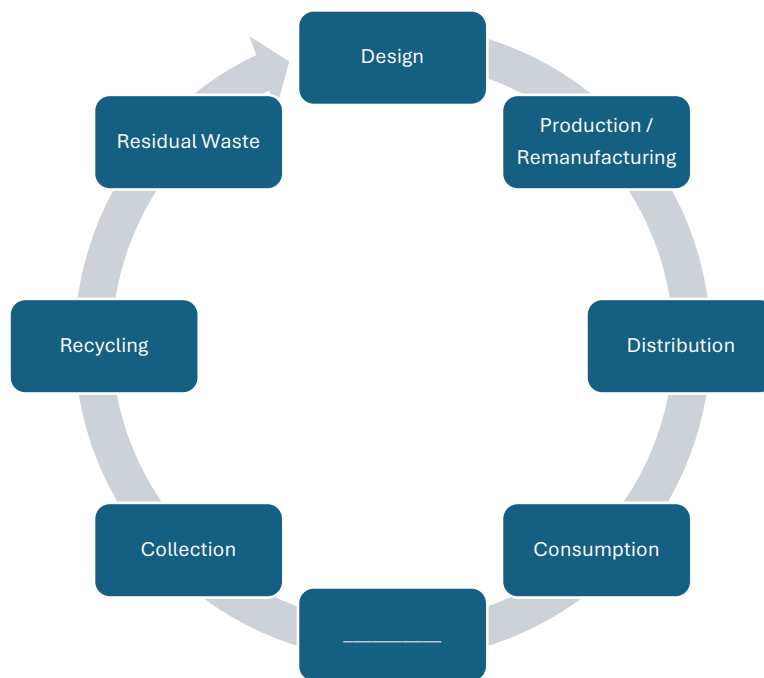


Figure 1: A circular economy model

Figure 1 shows a circular economy model with one stage left unlabelled. Which stage is missing?

- A. Landfill disposal
- B. Extraction of virgin raw materials
- C. Planned obsolescence
- D. Repair and reuse

2. Stanley 1913 has manufactured vacuum-insulated steel drinkware since 1913. Its *Quencher* travel tumbler, first sold in 2016, became a social media phenomenon and sales grew rapidly.

The Quencher has a double-walled stainless steel body with a vacuum between the walls, a moulded plastic handle, a tapered base and a resealable lid with an opening for a straw. It is released in new colours several times a year.



Figure 2: The Stanley Quencher tumbler, in all the colors of the saddest rainbow (Stanley.com, accessed 06-09-2026)

Which physical property is the designer relying on when specifying the material for the inner and outer walls of the Quencher, see Figure 2?

- A. Low density
- B. High melting point
- C. Low thermal conductivity
- D. High electrical resistivity

3. The base of the Quencher is tapered so that the tumbler fits inside a standard car cup holder, see Figure 2.

Which ergonomic consideration is being applied?

- A. Clearance
- B. Reach
- C. Work envelope
- D. Range of sizes

4. Many Quencher owners buy an additional tumbler in a new seasonal colour while the one they already own still works perfectly.

Which type of obsolescence does this best illustrate?

- A. Functional obsolescence
- B. Style obsolescence
- C. Technological obsolescence
- D. Planned obsolescence

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5. A reusable tumbler replaces single-use cups, but a household that owns six Quenchers has consumed more steel and plastic than one that owns a single tumbler.

Which statement best describes the conflict for Stanley between the three Ps of the triple bottom line (TBL)?

- A. People benefit from lower prices while profit increases
- B. Planet benefits from recyclable steel while people benefit from a lower purchase price
- C. Profit from repeat colour releases conflicts with the planet cost of extra production
- D. Profit and people are aligned because reusable products always reduce environmental impact

6. Bambu Lab manufactures desktop 3D printers for consumers and small design studios. The X1 Carbon (Figure 3) has an enclosed heated chamber, a gantry made from carbon-fibre-reinforced polymer, and an automatic material system that holds four filament spools at once.

The printer builds a part by extruding heated thermoplastic filament through a moving nozzle onto a build plate, one layer at a time. A design studio can print a prototype overnight instead of waiting days for an external service.

Bambu Lab also runs MakerWorld, an online platform where users publish and download printable models.



Figure 3: The Bambu Lab X1 Carbon printer, with the automatic material system mounted above (Taken by Mr. K, 06-09-2026)

(a) Which rapid prototyping technology does the X1 Carbon use, see Figure 3?

- A. Stereolithography (SLA)
- B. Selective laser sintering (SLS)
- C. Computer numerical control (CNC)
- D. Fused deposition modelling (FDM)

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(Question 6 continued)

(b) A design team prints a full-size model of a handheld device in the correct colour and surface finish. The model has no working parts, and is shown to focus group participants to gather reactions to its appearance.

Which type of prototype have they produced?

- A. High-fidelity aesthetic prototype
- B. Low-fidelity aesthetic prototype
- C. High-fidelity functional prototype
- D. Low-fidelity functional prototype

(c) The X1 Carbon's moving gantry is made from a carbon-fibre-reinforced polymer rather than from the polymer alone, see Figure 3.

Which property is most improved by combining the two materials?

- A. Density
- B. Thermal conductivity
- C. Stiffness
- D. Corrosion resistance

(d) A team prints three versions of a handle at different diameters, asks users to grip each one, changes the design in response and prints again.

Which design process activity does this describe?

- A. Secondary research
- B. The model, test, refine cycle
- C. A task analysis
- D. A SWOT analysis

(e) A user whose product has a broken plastic clip downloads a replacement clip from MakerWorld and prints it, rather than discarding the product.

Which circular economy strategy does this most directly support?

- A. Dematerialization
- B. Take-back legislation
- C. Use of renewable energy
- D. Repair

7. Pokémon GO is a mobile game, released in 2016, that places game characters into a live camera view of the player's surroundings using the phone's camera and GPS. Players walk to real locations to find characters and to reach "PokéStops" positioned at landmarks, public artworks and buildings.

Within two months of release the game had been downloaded more than 500 million times.

Reports of players walking into traffic and entering private property led the developer, Niantic, to add a warning screen at every launch and to disable gameplay above a set travel speed.



Figure 4: The exciting of Pokémon GO (BBC.com, Accessed 07-09-2026)

(a) Which modelling technology is the basis of the gameplay shown in Figure 4?

- A. Augmented reality (AR)
- B. Virtual reality (VR)
- C. Motion capture
- D. Finite element analysis (FEA)

(b) Niantic added a launch warning screen and disabled gameplay above a set travel speed.

Which responsibility of the designer is being met?

- A. Ensuring the product is affordable
- B. Ensuring the product is safe to use
- C. Ensuring the product is recyclable
- D. Ensuring the product is upgradable

(This question continues on the following page)

(Question 7 continued)

(c) A new player can open Pokémon GO for the first time and catch a character within a minute, without reading any instructions.

Which usability objective does this demonstrate?

- A. Efficiency
- B. Learnability
- C. Memorability
- D. Satisfaction

(d) Some game information is shown as small coloured icons on a phone screen held in bright sunlight, see Figure 4. Players report difficulty telling the icons apart.

Which limiting aspect of user capability is most relevant?

- A. Hearing thresholds
- B. Muscle control
- C. Colour perception
- D. Fatigue

(e) Niantic's designers want to understand how players actually behave at a busy PokéStop: the routes they take to reach it, where they stand and how long they stay.

Which user-centred research method is most appropriate?

- A. User observation
- B. A survey
- C. A focus group
- D. A literature search

Questions continue on the next page.

8. The Eames Lounge Chair (Figure 5) was designed by Charles and Ray Eames in 1956 and has been manufactured continuously ever since. Its seat and back are curved shells of thin wood veneers bonded together and pressed into shape. The cushions are leather over foam, and the chair sits on a cast aluminium base.

Chairs made in 1956 are still in daily use. Herman Miller offers the chair in walnut, santos palisander and white ash, with a choice of leather colours, but has not changed its form.



Figure 5: An Eames Lounge Chair and ottoman with plywood shell and cast aluminium base. (Dextraliving.com, Accessed 07-09-2026)

(a) Which structural type best describes the chair as a whole, see Figure 5?

- A. Frame structure
- B. Combination structure
- C. Shell structure
- D. Solid structure

(b) The Eames Lounge Chair has remained desirable and in production for almost seventy years.

Which form of obsolescence has the design most successfully resisted?

- A. Functional obsolescence
- B. Technological obsolescence
- C. Planned obsolescence
- D. Style obsolescence

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(Question 8 continued)

(c) Herman Miller varies the chair's veneer and leather while leaving its form unchanged, see Figure 5.

Which basis for material selection is being varied?

- A. Mechanical properties
- B. Chemical properties
- C. Aesthetic characteristics
- D. Availability

9. The Sega Dreamcast (Figure 6) was released in 1998. It was the first home games console sold with a built-in modem for online play, and its memory card carried a small screen and buttons of its own.

Sega discontinued the Dreamcast in 2001, after competing manufacturers released consoles with DVD drives and much larger software libraries. Reviewers praised the Dreamcast's hardware at the time, and a community of enthusiasts still develops new games for it.



Figure 6: A Sega Dreamcast console and its memory card with integrated screen (en.wikipedia.org, Accessed 07-09-2026)

(a) Sega withdrew the Dreamcast after competing consoles offered DVD playback and larger game libraries.

Which type of obsolescence does this best illustrate?

- A. Technological obsolescence
- B. Style obsolescence
- C. Planned obsolescence
- D. Social obsolescence

(Question 9 continues on next page.)

(Question 9, continued)

(b) A SWOT analysis of the Dreamcast is carried out in 1999. The imminent release of competing consoles with DVD drives is recorded.

Under which heading does it belong?

- A. Strength
- B. Weakness
- C. Opportunity
- D. Threat

(c) A designer is asked to explain why the Dreamcast failed commercially despite strong reviews.

Which research approach is most appropriate?

- A. Primary research using material testing
- B. Primary research using a focus group of current designers
- C. Secondary research using sales figures and reviews published at the time
- D. Secondary research using the console's assembly drawings

10. Cotopaxi is an outdoor equipment company founded in 2014 and certified as a B Corporation, meaning it is independently assessed against social and environmental standards as well as financial ones.

Its Del Día range of backpacks (Figure 7) is made from remnant fabric: offcuts and part-used rolls bought from other manufacturers, which would otherwise be discarded. Each pack is assembled by a sewing operator who chooses the colour combination, so no two packs are alike.

The company publishes the weight of remnant material it diverts from landfill each year and donates a percentage of revenue to poverty relief.



Figure 7: Six Cotopaxi Del Día backpacks in a different combination of remnant fabric colours (Cotopaxi.com, Accessed 07-09-2026)

(a) Cotopaxi buys offcuts and part-used rolls from other manufacturers to make the Del Día range, see Figure 7.

Which circular economy objective is being met?

- A. Powering production with renewable energy
- B. Eliminating waste from another manufacturer's process
- C. Designing the pack so that it can be upgraded later
- D. Introducing take-back legislation

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(Question 10 continued)

(b) Cotopaxi's B Corporation certification requires it to report on social and environmental performance alongside financial performance.

Which framework does this most closely reflect?

- A. The triple bottom line (TBL)
- B. Datchefski's five principles
- C. A SWOT analysis
- D. The five usability objectives

(c) Del Día packs are made from a ripstop nylon, a fabric woven with a reinforcing grid of thicker threads so that a small cut does not spread across the panel.

Which mechanical property does the ripstop weave most improve?

- A. Hardness
- B. Elasticity
- C. Ductility
- D. Toughness

(d) Because remnant fabric is bought in whatever quantities happen to become available, Cotopaxi cannot promise that any particular colour can be supplied again.

Which factor in material selection does this constraint represent?

- A. Cost
- B. Availability
- C. Aesthetic characteristics
- D. Mechanical properties

(Question 10 continues on next page.)

(Question 10 continued)



Figure 8: The Allpa (non-Del Dia variant) shoulder straps/harness (Cotopaxi.com, Accessed 07-09-2026)

(e) The Del Día pack is sold in a single size, with shoulder straps and a sternum strap that move over a wide range, see Figure 8.

Which strategy has Cotopaxi used to fit a wide percentile range?

- A. Designing the product to be adjustable
- B. Producing the product in a range of sizes
- C. Designing for the 50th percentile only
- D. Designing for the 5th percentile only

(f) The designer must set the maximum length of the adjustable shoulder straps, see Figure 8.

Which anthropometric percentile should be used?

- A. The 5th percentile
- B. The 50th percentile
- C. The 95th percentile
- D. The 5th–95th percentile range

(Question 10 continues on next page.)

(Question 10 continued)

(g) Cotopaxi tests the pack's buckles and zip pulls by asking users wearing thick winter gloves to open and close them.

Which inclusive design strategy is being applied?

- A. Designing for the average person
- B. Designing for a single primary persona
- C. Designing for the 50th percentile child
- D. Designing for extremes

	Maya, 24 Studies at a city university. Commutes 6 km by bicycle each day and hikes at weekends. Wants one bag that works for both. Carries a laptop, a waterproof and lunch.
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Figure 9a: A "real" Cotopaxi user profile. (image from Cotopaxi.com, Accessed 07-09-2026)

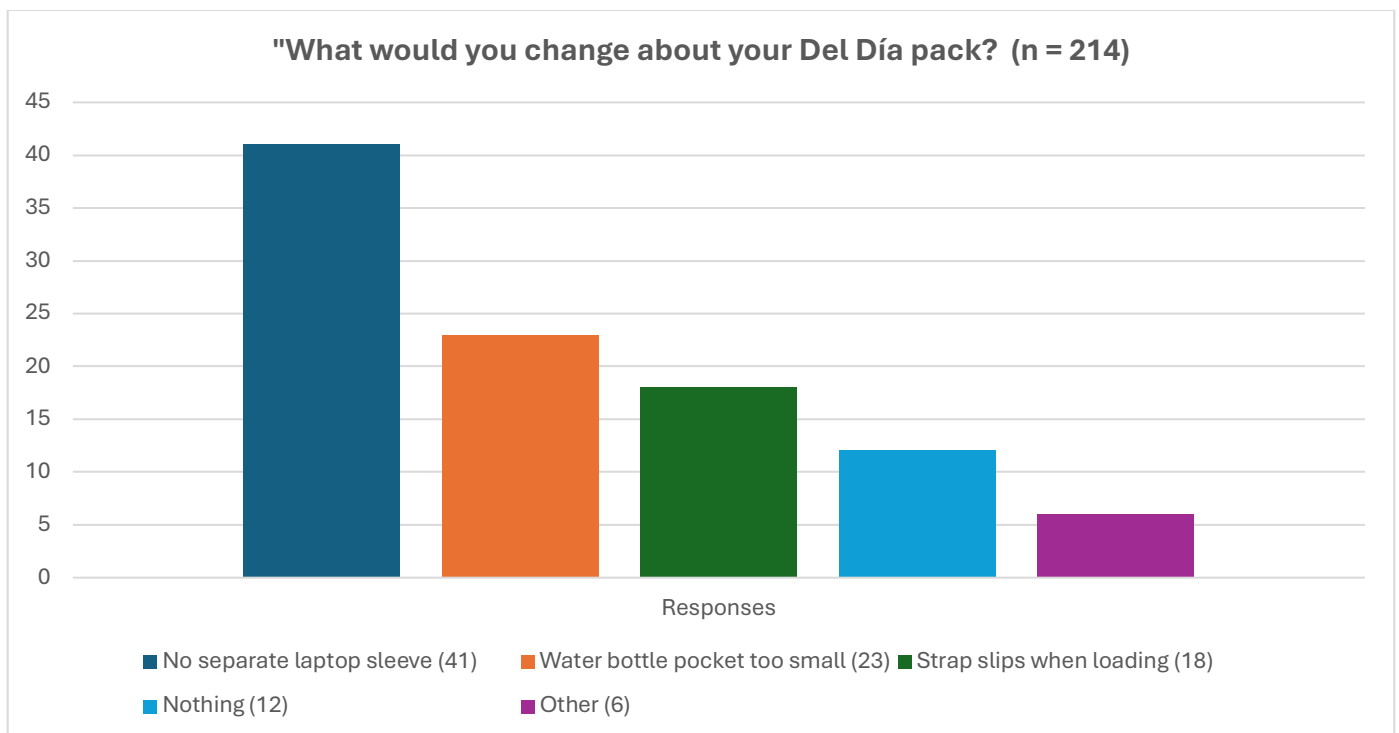


Figure 9b: Extracts from "real" Cotopaxi user research

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(Question 10 continued)

(h) Cotopaxi's research team combines its interview and survey data into the single profile shown in Figure 9a.

What is this profile called?

- A. A scenario
- B. A storyboard
- C. A primary persona
- D. A task analysis

(i) The bar chart in Figure 9 shows that the most common thing users would change is the absence of a separate laptop compartment.

Which product analysis concept does this feedback represent?

- A. Constructive discontent
- B. Reverse engineering
- C. Dematerialization
- D. Planned obsolescence

Markscheme

Design technology Standard level Paper 1 Set A

1.	D	7a.	A	9c.	C
2.	C	7b.	B	10a.	B
3.	A	7c.	B	10b.	A
4.	B	7d.	C	10c.	D
5.	C	7e.	A	10d.	B
6a.	D	8a.	B	10e.	A
6b.	A	8b.	D	10f.	C
6c.	C	8c.	C	10g.	D
6d.	B	9a.	A	10h.	C
6e.	D	9b.	D	10i.	A
